

AADITYA N SANIL

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Computer Science

Expected May 2028

- GPA: **3.70** | **Relevant Coursework:** Discrete Math, Data Structures, Principles of Data Science, Optimization Models in Engineering, Computer Architecture, Probability

EXPERIENCE

Martin Computational Speciation Lab

Jun. 2025 – Present

Undergraduate Researcher

Berkeley, CA

- Reduced simulation latency by **83%** (60m $\rightarrow$ 10m) by refactoring the **Julia** backend for parallel HPC SAVIO cluster
- Architected a **Distributed Asynchronous Population Engine** to generate and evaluate **100k+ concurrent entities** across multiple nodes for high-fidelity trait evolution.
- Engineered an automated telemetry pipeline to parse and filter **26 performance metrics** per run, enabling rigorous regression testing and experiment analysis.

Imagine Games Network (IGN)

Aug. 2025 – Dec. 2025

Software Engineer Intern (Contract)

Los Angeles, CA

- Engineered feature selection microservices to process **20k+ sparse signals**, utilizing **Lasso Regression** to optimize input layers for cognitive AI models.
- Optimized heap utilization and ingestion workflows using **Polars** to enable stable, high-performance processing of **500k+ user records**.
- Architected **FastAPI** microservices with real-time telemetry and **GitHub Actions CI/CD** to automate audience segment activation for a production-ready launch of Imagine AI

DreamWorks Animation

Feb. 2025 – May 2025

Software Engineer (Contract)

Glendale, CA

- Refactored **MoonRay MCRT** rendering kernels to optimize production pipeline runtimes via memory alignment.
- Engineered **Python automation scripts** to bridge manual storyboarding with LLM backends, accelerating pre-production cycles.
- Developed **inference latency and memory benchmarks** to evaluate the technical feasibility of AI integration across studio pipelines.

PROJECTS

SynLlama: LLM-based Synthesis Planning | *Python, PyTorch, Llama-3, SMILES*

- Assisted in **planning and implementation** for SynLlama, a Llama-3 framework.
- Developed a **high-efficiency fine-tuning pipeline** using reaction fragmentation, achieving a **96.4% success rate** with **100x less training data**.
- Validated model performance in **Enamine REAL space** (38B+ compounds), yielding optimized leads for 5-step synthetic routes.

Autonomous Financial AI Agent | *FastAPI, RAG, Alpaca API* | [GitHub](#)

Jan. 2026 – Present

- Architected a **hybrid RAG pipeline** to ingest unstructured SEC filings into an in-memory vector index for evidence-based LLM reasoning.
- Engineered a **risk-gated execution engine** via the Alpaca API, featuring a real-time observability dashboard for portfolio tracking.
- Achieved **90%+ core logic test coverage** by developing an automated evaluation framework to measure retrieval accuracy via **Pytest**.

AI Security: Provably Robust Classifiers | *PyTorch, CVXPY* | [GitHub](#)

Jan. 2026 – Apr. 2026

- Simulated **FGSM adversarial attacks** to identify deep learning vulnerabilities and measure classification degradation.
- Architected a **formal verification framework** using MILP and Lagrangian convexity to generate mathematical robustness certificates.
- Engineered an optimization pipeline using **CVXPY** to balance model accuracy against adversarial resilience during training.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Julia, SQL, R, Bash

Machine Learning & Data: PyTorch, Scikit-learn, Polars, NumPy, Pandas, CVXPY, DuckDB

Backend & Distributed Systems: FastAPI, Flask, PostgreSQL, Docker, HPC (SLURM/SAVIO), RESTful APIs

Tools & Infrastructure: Git, Linux/Bash, GitHub Actions (CI/CD), JWT/OAuth, Firebase, Render